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Task Description:	Abstract: Astronauts recovering from prolonged exposure to microgravity exhibit a striking similarity to acute vestibular deficit subjects. Theoretically, the brain is not receiving adequate information from the vestibular otolith system in either group: in astronauts, this is because of the absence of gravity, while in patients it results from receptor damage. Astronauts recover (adapt) but many vestibular patients do not recover and/or cannot adapt. By comparing responses to linear acceleration (of which gravity is a type) in the two groups, similarities and differences in mechanisms of adaptation to changing gravitational environments can be quantified. We hypothesize that reference to linear acceleration calibrates the use of the more ambiguous orientation references (vision and somatosensory) for sensorimotor adaptive control. This study will establish a population of vestibular deficient subjects with mild to severe loss of vestibular function as a ground-based model for the physiological effects of loss of gravity in space. Centrifuges and other rotation devices will be used to generate linear acceleration (artificial gravity). Perceptual, eye movement, and postural responses		

	will be used to study changes in the orientation and magnitude of linear acceleration on the vestibular and converging physiological systems. Results will be used to design rehabilitation methods to speed recovery from microgravity and to develop artificial gravity countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Vestibular disorders are very common, affecting approximately 90 million Americans (three times as many as hearing loss) over their lifetime. Loss of work and disability due to vertigo, imbalance, and spatial disorientation is very costly for both the afflicted individual and society. Vestibular disorders are a common cause (estimates as high as 70-80%) for falls in the elderly. Spatial disorientation-related accidents result in significant loss of life and equipment (cost estimates approximately $300 million/year) in the military each year. This research seeks to develop new therapeutics or protocols for alleviating vestibular diseases and disorders on earth, including Meniere's disease, ataxia (including mal de debarquement following cruises which are very disabling for the elderly), recovery from acute vestibular insults, avoidance of (and physiological compensation for) ototoxicity, and common motion sickness. NASA-sponsored research has contributed to diagnostic methods and treatment of these debilitating problems on Earth. These results also have enabled the development of effective, efficient clinical rehabilitation protocols. NASA research has also, in part, sponsored research that resulted in the most widely used method for documentation of postural instability and outcome of therapy (computerized dynamic posturography). The EquiTest postural control (computerized dynamic posturography) assessment system developed, in part as a consequence to NASA sponsored research) is now the world standard for assessment and quantification of vestibulospinal disorders and postural instability. The EquiTest® system is used to assess candidates for Navy flight programs, and has become a part of the routine medical assessment of returning Space Station astronauts. We anticipate that future developments will assist clinicians in the selection and monitoring of patients undergoing treatment regimens and may detect persons at risk for falls in the aging. The mechanisms of sensorimotor adaptive control of human posture and movement are being characterized by our research. For example, we have characterized the minimal vestibular input, as a function of frequency and amplitude, required for dynamic visual stabilization in the vestibular deficient human. Results of NASA-sponsored vestibular research have produced the only peer reviewed, large N, published normal databases and have contributed significantly to our understanding of vestibular adaptive and compensation processes. The only environment in which environmental inputs to the otolith vestibular system can be removed (or systematically reduced) and re-introduced systematically is in the microgravity environment of space. The fundamental conditions, in the strict sense, for scientific investigation of vestibular mechanisms therefore require the microgravity environment of space flight. Our work expands upon the classical studies of vestibular mechanisms and, through collaboration with NASA scientists and other colleagues performing ground-based research, provides the terrestrial database (and technology) for support of space research in humans. Our laboratory published the first large N database of vestibular function tests (both vestibulo-ocular and vestibulospinal) in the same human subject, and have tested some of those same subjects a decade later for the first longitudinal study performed in humans. Results of this study show that the only significant changes occur under conditions that require vestibular input for control of posture. This finding supports the hypothesis that changes in postural control, at least in the age range 20-80 years, postural instability is related to increased probability of vestibular disease and is not related to aging per se. Our team has developed a new surgical technique for the repair of perilymph fistulas (fluid leak from the inner to the middle ear). This is the first known surgical treatment for motion sickness associated with a vestibular disorder.
	Summary of Progress: Normal subject responses to varying linear accelerations (magnitude and direction, visual feedback and verbal estimates of tilt/translation is frequency dependent. Frequency dependent variability in responses was especially large when the protocols involved LVOR-VOR interactions using perceptual tasks. Based upon results of our studies to date we suspect that the perceptual responses vary as a function of canal-otolith interactions. We conclude that the frequency-dependent properties of canal-otolith interactions and their perceptual dynamics determine adaptive responses to novel force environments in both normal and vestibular abnormals. Adaptation experiments must therefore define and control for frequency-dependent visual-vestibular and somatosensory-vestibular interactions. We have also demonstrated that "double integration" of x-y linear acceleration distributed as an array to the anterior tongue (Bach-y-Rita's "Brain Port" technique) re-instates the ability of the brain to interact ambiguous visual and somatosensory inputs for control of upright stance. The effect transfers to active movements including gait and exhibits retention lasting hours to days. These results obviously have very important implications for effective countermeasures for the untoward effects of microgravity on neuromuscular control. The purpose of this study was to evaluate the contribution of canal and otolith inputs in resolving linear accelerations arising from roll tilt and/or lateral translation as a function of stimulus frequency. Einstein's equivalence principle states that there is an ambiguity between linear accelerations arising from tilts with respect to gravity versus translational motions. During tilt, the semicircular canals detect rotational motion, while the otoliths detect head orientation with respect to gravity. Both confirm the magnitude of tilt motion. During a translational motion in the opposite direction, the resultant force vector is a function of both gravity and translational acceleration. If the subject perceived the resultant force to be down, the same tilt perception would result. Two proposed neural strategies used to resolve this ambiguity are 1) frequency segregation and 2) multisensory integration. We hypothesize that the absence of a gravity vector during spaceflight leads to adaptive changes in these neural strategies used for resolving ambiguous linear accelerations. If so, understanding these neural processes is essential for the development of effective countermeasures. Eight normal subjects were exposed to sinusoidal linear acceleration stimuli during two different sessions. Roll tilt amplitude of ±20 deg was maintained across frequencies (0.01 and 1.0 Hz). Session 1: roll tilt stimuli were provided during rotation about the naso-occipital axis at eye level (EHA). Session 2: subjects were exposed to linear acceleration profiles using a combination of translation and centrifugation (VRC), except for 1.0 Hz stimuli that involved only translation (±0.09 m). Stimulus order and session order were counterbalanced. Ocular responses were recorded using infrared video-oculography, and perceived tilt and translation were evaluated using verbal reports and continuous settings of a tilt bar that allowed quantification of both magnitude and phase of roll tilt perception.

Task Progress:	Results: (1) Verbally reported perceptual estimates of tilt and translation did not differ between naso-occipital roll-tilt (EHA) and perceived responses to linear acceleration (VRC) at low frequencies (<0.1 Hz). At 0.01 Hz, subjects overestimated magnitude of tilt by a factor of two. At higher stimulus frequencies (>0.1 Hz) during naso-occipital rotation, subjects continued to report the perception of roll-tilt without translation. At higher stimulus frequencies during linear acceleration without canal cues, the perception of roll-tilt was replaced by sensation of translation. At 1 Hz linear acceleration, subjects perceptually over-estimated the actual translation by an order of magnitude (perceived ± 1m vs ~ ± 10 cm actual). (2) Somatosensory estimates of earth horizontal (indicated by setting of a tilt bar to perceived earth horizontal) were similar to but more accurate (less variable) than verbal reports of tilt magnitude. There was no difference between naso-occipital roll-tilt (EHA) and linear translation (VRC) bar settings at low frequency (0.01 Hz). Subjects made accurate settings of the bar along the horizon at higher stimulus frequencies during EHA rotation when both canal and otolith cues were present. During VRC conditions when canal cues were absent, the magnitude of tilt perception as indicated by the bar settings decreased as a function of frequency, with negligible sense of tilt at 1 Hz. Results (1) and (2) suggest that canal cues are important for tilt detection above 0.1 Hz, but do not contribute significantly to tilt perception re: earth horizontal below 0.01 Hz. Findings are consistent with Nashner, et al. which predicts that canal inputs are effective for dynamic tilt detection only at frequencies >0.1 Hz(1) (3) Tilt perception estimated by somatosensory bar settings to perceived earth horizontal was accurate when the peak tilt velocity was maintained at ±38 deg/sec across a frequency range of 0.1 to 1 Hz. Tilt perception also did not differ during centrifugal linear acceleration (VRC) conditions when the head was pitched forward, but tilt detection deteriorated as a function of frequency (compatible with a low pass filter model) in both head positions. EHA = naso-occipital (canal + otolith) roll-tilt. Results show that roll-tilt is not velocity dependent, at least at the magnitudes studied. (4) Ocular torsion slow phase velocity (SPV) represents the compensatory eye movement response to roll-tilt (canal + otolith). Error bars = SEM. During centrifugal linear acceleration (VRC) without canal (otolith only) cues, the magnitude of the torsional SPV decreased as a function of stimulus frequency. As with tilt perception, the torsional SPV response did not differ significantly when the head was pitched forward during VRC. (5) When evaluating the response of horizontal eye movement slow phase velocity (SPV) (compensatory eye movement responses) to lateral translation (centrifugal linear acceleration) as a function of frequency, horizontal SPV increased as a function of stimulus frequency. This was in contrast to the torsional SPV response. This effect was also present during naso-occipital (EHA) rotation. While reducing the stimulus amplitude at 1 Hz from ±126°/s to ±38°/s had a tendency to increase the torsional SPV gain, it had the opposite effect for horizontal SPV as seen here. The horizontal SPV response at this higher frequency is more robust during EHA conditions when the tilt velocity amplitude is greater. (6) Perception of roll tilt and modulation of ocular torsion responses obtained from naso-occipital roll-tilt and linear acceleration were similar at 0.01 Hz. There was negligible perception of translation and modulation of horizontal SPV at 0.01 Hz. As the stimulus frequency increased during naso-occipital rotation when both canal and otolith cues were present, modulation of torsion and perception of roll tilt was evident. However, as stimulus frequency increased during variable radius centrifugation (linear acceleration/translation only) when primarily otolith cues were presented, both the modulation of torsion and perception of roll-tilt decreased while modulation of horizontal SPV and perception of translation increased. These studies document the frequency-dependent perceptual and eye movement responses to roll-tilt and linear acceleration in the human, and permit us to generate stimulus protocols for the systematic study of the contributions of canal and otolith inputs to body segment control in normal and abnormal subjects. Results of this study were presented in part at the 2003 Bioastronautics meeting in Galveston, TX. Papers have been submitted for publication. (7) Using a third paradigm, subjects were sinusoidally roll tilted but with ears above or below the rotation axis. This paradigm provided sinusoidal roll canal cues that were the same across trials while providing sinusoidal otolith cues that varied linearly with ear position distance from the axis of rotation. The data clearly showed that the perception of tilt and perception of translation were dependent on canal cues, with substantial roll tilt and inter-aural translation perceptions reported even when no inter-aural force cues were present. The above findings collectively match internal model predictions that rotational cues from the semicircular canals influence the neural processing of otolith cues. The data also demonstrated horizontal translational VORs at higher frequencies (>0.2Hz) during both Translation and Tilt; these responses were dependent on otolith cues but independent of canal cues. These findings match frequency segregation predictions that translational VORs include contributions via simple high-pass filtering of otolith cues. More generally, these findings demonstrate that internal models govern human perception while showing that simple filtering contributes substantially to the human translational VOR at frequencies above 0.2 Hz. We propose to continue these studies in subjects with stable, well documented unilateral and bilateral vestibular function loss as a ground based population model for adaptation to microgravity.
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